Vietnamese-German University

Khoa Cai Phuc Thien <khoa.cpt@vgu.edu.vn>

Urgent need of your help with two questions from OFIN students

Ngoc Doan Minh <10322013@student.vgu.edu.vn> Thu, Jul 11, 2024 at 3:20 PM
To: Khoa Cai Phuc Thien <khoa.cpt@vgu.edu.vn>

Dear Mr. Khoa,

Thank you for your email.

Best regards,
Ngoc Doan

On Thu, 11 Jul 2024 at 15:03 Khoa Cai Phuc Thien <khoa.cpt@vgu.edu.vn> wrote:

> Hi Ngoc and Anh,
>
> Please see Dr. Omer's replies below.
>
> Best,
> Khoa Cai
>
> ---------- Forwarded message ---------
> From: <omer.cayirli@vgu.edu.vn>
> Date: Thu, Jul 11, 2024 at 2:57 PM
> Subject: RE: Urgent need of your help with two questions from OFIN students
> To: Khoa Cai Phuc Thien <khoa.cpt@vgu.edu.vn>
>
> Khoa,
>
> When there are no taxes r(u), r(a) and r(WACC) are equal.
>
> *"(b) Is **the value of the plant higher** than one determined in (a) because of the reduction in energy cost from $8.75mil to $3mil? FCFnew = (50 - (50 x 0.7 - 8.75) - 3) x (1 - T)?"*
>
> **The value increases b/c of a significant increase in FCF as a result of the reduction in energy costs.**
>
> *"(c) Will the **plant's cost of capital decrease** because the long-term contract helps remove the energy cost fluctuations, and lower the cost to only $3 mill / year."*
>
> **The cost of capital does not decrease. Instead, it increases to 9.27% from 7.95%. That is because of increased risk as a result of fixed costs (Operational leverage).**
>
> ---
>
> **From:** Khoa Cai Phuc Thien <khoa.cpt@vgu.edu.vn>
> **Sent:** Thursday, July 11, 2024 9:52 AM
> **To:** Omer Cayirli <omer.cayirli@vgu.edu.vn>
> **Subject:** Re: Urgent need of your help with two questions from OFIN students

Dear Dr. Omer,

A student came to the tutorial session today and asked me two questions in Exercise set 08, I gave her the answers to some parts of them, but need your help with the remaining:

Question 2: Consider the following airline industry data from....

(b). Estimate the asset beta for each firm. How to calculate asset beta in this case without tax rate?

Question 4: Your company operates a steel plant...

(b) Is **the value of the plant higher** than one determined in (a) because of the reduction in energy cost from $8.75mil to $3mil? FCFnew = (50 - (50 x 0.7 - 8.75) - 3) x (1 - T)?

(c) Will the **plant's cost of capital decrease** because the long-term contract helps remove the energy cost fluctuations, and lower the cost to only $3 mill / year.

Best,

Khoa Cai

On Wed, Jul 10, 2024 at 9:42 PM Khoa Cai Phuc Thien <khoa.cpt@vgu.edu.vn> wrote:

> Okay, thank you, Dr. Omer.
>
> Best,
>
> Khoa Cai
>
> On Wed, Jul 10, 2024 at 9:39 PM Omer Cayirli <omer_cayirli@uncbusiness.net> wrote:
>
> > It is always stated in questions if continuous compounding needed. If not, they may use discreet approach. Not a big deal at OFIn level
> >
> > On Wed, Jul 10, 2024, 5:23 PM Khoa Cai Phuc Thien <khoa.cpt@vgu.edu.vn> wrote:
> >
> > > Dear Dr. Omer,
> > >
> > > A student has recently asked me when he needs to make an assumption about "continuously compounded" with forward price questions? In this situation, I cannot give him a reliable answer.
> > >
> > > This is a solution to a question in a given exercise set he found on the Internet:

, and he wondered why the continuously compounded formula was used.

Best,

Khoa Cai

On Wed, Jul 10, 2024 at 10:54 AM Khoa Cai Phuc Thien <khoa.cpt@vgu.edu.vn> wrote:

> Okay, thank you Dr. Omer.
>
> Best,
>
> Khoa Cai

On Wed, Jul 10, 2024 at 10:53 AM Omer Cayirli <omer_cayirli@uncbusiness.net> wrote:

> Yes. They should read the question carefully.

On Wed, Jul 10, 2024, 6:49 AM Khoa Cai Phuc Thien <khoa.cpt@vgu.edu.vn> wrote:

> Dear Dr. Omer,
>
> Yesterday, the student also concerned about the increase in Net Working Capital.
>
> As a new project requirement, initial investment in net working capital in year 0, and/or the increases in net working capital in each year of the project's lifetime normally occur.
>
> If a question does not give any information or say anything about the recovery of those additional net working capital in the final year, could we ignore or not take into account the recovery?
>
> Best,
>
> Khoa Cai

On Wed, Jul 10, 2024 at 9:51 AM Omer Cayirli <omer_cayirli@uncbusiness.net> wrote:

> 👍

On Wed, Jul 10, 2024, 5:50 AM Khoa Cai Phuc Thien <khoa.cpt@vgu.edu.vn> wrote:

> Dear Dr. Omer,

Thank you so much for your reply.

Regarding the revenue growth rate issue, at first I just looked at her excel file, after your reply I have double checked it by my own excel calculation, and found out that number of **members x membership fee** (direct) and **revenue x (1 + 9.18%)** (indirect) produced exactly the same result as the slide.

I am going to give her your explanations in today's tutorial sessions.

Best,

Khoa Cai

On Wed, Jul 10, 2024 at 12:55 AM <omer_cayirli@uncbusiness.net> wrote:

> Dear Khoa,
>
> I cannot figure out why APR = 3.78/12 is not used for PV(port)(real) calculation instead of EAR = 3.85%/12 = 0.003208?
>
> *You are correct. It should be APR/12. Although it is 0.003208, the actual calculations and the result are correct. Using APR/12, you should get the result on the slide.*
>
> The student calculated revenues directly by multiplying number of members and membership fee in consideration of their growth rates of 3% and 6% respectively.
>
> Her result is about 1,600,000 which is different from 1,373,279.31.
>
> *I don't know how she got the 1,600,000 number. If she had done it correctly, she should have had the sum of the 4th column(revenue), which is 2,116,123.89. There is no problem with the solution. The only difference can be the interpretation of "puts aside the amount he will spend in a year at the start of that year", nothing else.*
>
> Let me know if you need further help.
>
> Omer.
>
> ---
>
> **From:** Khoa Cai Phuc Thien <khoa.cpt@vgu.edu.vn>
> **Sent:** Tuesday, July 9, 2024 7:21 PM
> **To:** Omer Cayirli <omer_cayirli@uncbusiness.net>
> **Cc:** Khoa Cai Phuc Thien <khoa.cpt@vgu.edu.vn>
> **Subject:** Urgent need of your help with two questions from OFIN students
>
> Dear Dr. Omer,

Could you please help me with the questions attached in this email?

Best regards,

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Kính thư / Mit freundlichen Grüßen / Kind Regards,

Cai Phuc Thien Khoa (Mr.)

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